CULTIVATION GUIDE | CROP MANAGEMENT

Paddy Fertiliser Timing and Field Inspection

Where a crop sits in the published fertiliser cycle, how the leaf colour chart is used, how rain changes timing, and what to inspect.

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Teaching edition for the STEMLink AI Engineer Bootcamp. Content restructured from AgriVibe knowledge base AV-KB-001 Rev 1.1, which grounds its agronomy in Department of Agriculture and RRDI guidance for Sri Lanka. Not for use with real farmers until the verification items in AV-POL-03 are closed.

SCOPE OF THIS GUIDE

This guide explains timing relative to weather and field condition, and where a crop sits within the published schedule. It does not give a quantity for any farmer's field. Quantities come only from the Grama Niladhari division recommendation or the local Agriculture Instructor (see AV-POL-02).

1. Why there is no single national dosage

The Department of Agriculture does not issue one national fertiliser figure for paddy. Recommendations vary by agro-ecological zone, the Wet Zone differing from the Intermediate and Dry Zone; by water regime, irrigated differing from rainfed; by varietal age class, three month, three and a half month or four month; and by location, with recommendations published at Grama Niladhari division level, district by district. Further field-specific refinements exist based on soil analysis reports, soil test kit results and leaf colour index values.

Because the recommendation is layered in this way, a farmer's correct quantity depends on information this system does not hold. That is the substantive reason a farmer is routed to the authoritative source, and it should be explained in those terms rather than as a bare refusal.

2. Where a crop sits in the schedule

The Department of Agriculture publishes four general schedules, one for each combination of zone (Wet Zone, or Intermediate and Dry Zone) and water regime (irrigated or rainfed). They are reproduced in AV-CG-06 with the districts each applies to. Every schedule times its applications from the date of establishment and by varietal age class, so a crop's age tells you which application it is approaching. For instance, a crop around forty days old is approaching the six week top dressing in a three month or three and a half month schedule. That makes the farmer's question one of timing, not quantity.

3. The leaf colour chart

The Department of Agriculture publishes fertiliser recommendations based on leaf colour index values, and the Rice Research and Development Institute (RRDI) uses the leaf colour chart in Sri Lanka. The chart is a small plastic card of graded green panels. The farmer compares the middle of the topmost fully expanded leaf against the panels, in the shade of their own body, without detaching the leaf, and repeats this across several plants in the field.

When a farmer reports yellowing and asks whether to apply urea, the leaf colour chart is the recognised field method for answering that question properly. Recommending that the farmer check the crop with a chart, or ask their Agriculture Instructor to do so, is a concrete and appropriate recommendation, and far more useful than either inventing a quantity or declining to help.

4. Nitrogen and heavy rainfall

Nitrogen should not be applied immediately before heavy rainfall. Nitrogen applied before heavy rain may be lost through runoff over the field bunds, particularly where surface drainage is poor. Water management and fertiliser application in rice are directly linked: the RRDI treats water control as part of fertiliser practice, and under its improved alternate wetting and drying (AWD) practice the water level in the field is managed

deliberately around applications.

Advice that fits this situation: consider waiting until forecast heavy rainfall has passed; check drainage, standing water and the condition of affected sections first; reassess once conditions have stabilised; and confirm the amount with the Grama Niladhari division recommendation or an Agriculture Instructor before applying.

5. Field inspection checklist

Where vegetation has declined and weather may be contributing, the following may be recommended for inspection. Select at most four items that fit the specific situation; do not reproduce the whole list.

- Standing or flowing water, and whether it is draining away.
- Blocked drainage, and whether interflow is entering from adjacent highland.
- Whether yellowing is widespread or concentrated in particular sections.
- Whether the discolouration is pale yellow, or reddish-brown and bronzed.
- Whether older or younger leaves are affected first.
- Whether plants are flattened or physically damaged.
- Whether weaker sections correspond with low-lying or poorly drained ground.
- When the field was last fertilised or irrigated.